

# Question Paper

Exam Date & Time: 31-Dec-2025 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026  
SUBJECT: ECE 1071/ECE\_1071 BASIC ELECTRONICS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) A PN-junction germanium diode has a saturation current of  $1\mu\text{A}$  for an applied forward voltage of  $0.2\text{V}$  at  $27^\circ\text{C}$ . Determine the static and dynamic resistance. Also calculate the reverse saturation current when temperature rises to  $47^\circ\text{C}$ . (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) With a neat diagram, explain the structure and operation of MOSFET. Also, sketch and explain the  $I_D$ - $V_{GS}$  and  $I_D$ - $V_{DS}$  characteristics. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 1C) With the help of an appropriate circuit, input & output waveforms, explain the working of a full wave center-tap rectifier with capacitor filter. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2A) A full wave Bridge rectifier circuit is fed by ac mains of rms  $230\text{ V}$ ,  $50\text{ Hz}$  supply through a transformer having turns ratio of  $10:1$ . If the load resistance is  $1\text{k}\Omega$ , assuming the diode resistance to be zero, determine (4)  
(i) Average output voltage.  
(ii) Average load current.  
(iii) Output DC power.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2B) In a voltage regulator using Zener diode, the output voltage  $V_0 = 5\text{V}$ , Input voltage  $V_i = (12 \pm 3)\text{ V}$ ,  $R_S = 1\text{k}\Omega$ , Load current  $I_L = 20\text{mA}$ . Find the minimum and maximum currents flowing through the Zener diode. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2C) Explain with a neat circuit diagram how op-amp works as differentiator and derive the expression for output voltage. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3A) Derive the expression for the output voltage of a non-inverting op-amp amplifier with a neat circuit diagram. Find the output voltage for input  $V_{in} = 2\sin(2\pi 100t)$ . Given  $\pm V_{CC} = 15\text{V}$ ,  $R_F = 10\text{k}\Omega$  and  $R_1 = 2\text{k}\Omega$ . Plot both input and output waveforms indicating all features in it. (4)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3B) A hexadecimal numbers  $B_{(16)}$  is transmitted using a (7,4) Hamming code with even parity. During transmission, an error occurs in the 3rd bit of the codeword. Analyse and determine what is the received codeword, showing all steps of reasoning. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 3C) A logic circuit compares 2 inputs (A and B). It produces 3 outputs G, L and E, where G is high when  $A > B$ , L is high when  $A < B$  and E is high when  $A = B$ . Design the circuit with the help of a truth table and a circuit diagram using basic gates. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4A) Using KMAP, Deduce the simplified Boolean expression and implement it using only NAND gates for the following expression. (4)

$$F = \sum_m (0, 2, 5, 7, 8, 13) + d(10, 15)$$

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4B) With aid of relevant circuit, truth table and timing diagram explain working of negative edge triggered JK flip-flop. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4C) Analyze the operation of a 4-bit Serial-In Parallel-Out (SIPO) shift register by interpreting its truth table and deducing its logical circuit for input data sequence of 1011. Explain how data propagates through the flip-flops with each clock pulse. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5A) Given an FM equation  $V_{FM}(t) = 10 \cos[2\pi 108t + 5 \sin(2\pi 15000t)]$ , Calculate Carrier frequency. Modulating frequency. Frequency deviation. Bandwidth using Carson's rule. (4)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 5B) Apply your understanding of communication systems to explain its components using a labeled block diagram. (3)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 5C) Given Binary information 10110101 plot ASK, FSK and PSK wave forms. (3)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

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| Event Name                        | Subject                    |
|-----------------------------------|----------------------------|
| 058-MIT BLR-MAKEUP EXAM-2025-2026 | ECE_1071:BASIC ELECTRONICS |

| # | Course Outcome | Description | Marks |
|---|----------------|-------------|-------|
| 1 | 1              | CO 1        | 4     |
| 2 | 2              | CO 2        | 13    |
| 3 | 3              | CO 3        | 23    |

\* **NA** - CO Not Selected

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|-----------------------------------|----------------------------|
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| # | Level       | No of Questions | Marks (%) |
|---|-------------|-----------------|-----------|
| 1 | Remember    | 0               | 0(0%)     |
| 2 | Understand  | 3               | 9(18%)    |
| 3 | Application | 0               | 0(0%)     |
| 4 | Analysis    | 9               | 31(62%)   |
| 5 | Create      | 3               | 10(20%)   |
| 6 | Evaluate    | 0               | 0(0%)     |
|   | Total       | 15              | 50 (100%) |